

Foundations of continuity equations for their neural applications

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Abstract

These notes lay foundational underpinnings for continuity equations and their roles in optimal transport (OT) problems that dominate the machine learning paradigm. These notes are based off a reading course with Professor Rongjie Lai at Purdue University. We will attempt to understand background relating to the continuity equation as a variational PDE and its relevant calculus of variations. We will connect these notes to existing recent and relevant literature, such as neural JKO algorithms and other recently-discovered phenomena, such as the Wasserstein geometry.

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1 JKO variational equivalence and the Lebesgue integral formulation

Consider the OT problem [Lee et al. \(2023\)](#)

$$\begin{cases} (\rho, v) = \arg \min \iint_{[0,1] \times \mathbb{R}^d} \rho \|v\|_2^2 dx dt + 2\Delta t \mathcal{E}(\rho(1, \cdot)) \\ \text{subject to } \partial_t \rho + \nabla(\rho v) = 0, \rho(0, x) \rho_0. \end{cases} \quad (1)$$

Equivalently, this can be written

$$\begin{cases} \min_v \iint_{[0,1] \times \mathbb{R}^d} \rho^n(x) \|v(T(t, x))\|_2^2 dx dt + 2\Delta t \mathcal{E}(T(1, \cdot) \# \rho^n) \\ \text{subject to } \frac{d}{dt} T = v, T(0, x) = x. \end{cases} \quad (2)$$

We discuss why these are equivalent. We already proved in section 8 why the Lagrangian flow map condition $\frac{d}{dt} T = v$ is equivalent to a continuity equation.

It is also known

$$\rho(\tau, \cdot) = T(\tau, \cdot) \# \rho^n. \quad (3)$$

There is a known equivalence [Villani](#) (chapter 1)

$$\int_Y \varphi(y) d\nu(y) = \int_X \varphi(T(x)) d\mu(x) \quad (4)$$

when $T \# \mu = \nu$. This implies $\rho(1, \cdot) = T(1, \cdot) \# \rho^n$. The result follows from these facts. In particular, we can take $\varphi = \|v\|_2^2$ and we see

$$\iint_{[0,1] \times \mathbb{R}^d} \rho(t, x) \|v(t, x)\|_2^2 dx dt = \iint_{[0,1] \times \mathbb{R}^d} \|v(t, x)\|_2^2 d\rho dt \quad (5)$$

$$= \iint_{[0,1] \times \mathbb{R}^d} \|v(t, T(t, x))\|_2^2 d\rho^n dt = \iint_{[0,1] \times \mathbb{R}^d} \rho^n(x) \|v(t, T(t, x))\|_2^2 dx dt. \quad (6)$$

Here, we have assumed our measures are absolutely continuous with respect to Lebesgue measure. In particular, it can be noted that Lebesgue measure coincides with Riemann integration if we are allowed the assumption of Riemann integrability, which we will indeed. As a reminder, we say a measure Γ is absolutely continuous with respect to Lebesgue measure λ^* , or $\Gamma \ll \lambda^*$, if there exists a density ρ such that

$$\Gamma(A) = \int_A f d\lambda^*, \quad (7)$$

where A is any measurable set on the Lebesgue σ -algebra.

2 Deriving the continuity equation

2.1 A more prototypical engineering derivation

Let $V \subseteq \mathbb{R}^d$ be a closed, connected, and bounded volume in Euclidean space and $\rho : \mathcal{V} \times [0, T] \rightarrow \mathbb{R}^+$ be a mass density. Let $v : \mathcal{V} \times [0, T] \rightarrow \mathbb{R}^d$ be a vector field. We have

$$\text{total mass} = \int_V \rho dV. \quad (8)$$

It is well known via flux theory in calculus that the change of mass through the surface is given by

$$\text{change of mass in the volume} = \int_{\partial V} \rho v \cdot n dS. \quad (9)$$

The mass is conserved in the volume, and we get

$$\partial_t \int_V \rho dV = - \int_{\partial V} \rho v \cdot n dS. \quad (10)$$

This is generally a well-known identity in engineering (see [Leishman \(2021\)](#)). Here, our surface integral is outward-oriented. We will assume basic regularity properties such as smoothness on ρ , and since our domain is sufficiently nice, we can exchange differentiation and integration. Hence,

$$\int_V \partial_t \rho dV + \int_{\partial V} \rho v \cdot n dS = 0. \quad (11)$$

We cannot combine the integrals into one equation yet due to the first being an integral over Euclidean space and the second a surface integral. Notice the criteria of the divergence theorem are satisfied (we will allow v to be sufficiently smooth), hence

$$\int_V \partial_t \rho dV + \int_V \nabla \cdot (\rho v) dV = \int_V (\partial_t \rho + \nabla \cdot (\rho v)) dV = 0. \quad (12)$$

Just because an integral is zero, that does not imply the integrand is zero. Moreover, ρ is not a (compactly supported) test function.

There are some ways to conclude the proof. For example, we have $\partial_t \rho + \nabla \cdot (\rho v)$ if we permit V to be arbitrary, and if the equation must hold for all V .

Let us justify the negative sign in equation 10 intuitively with a special case. Suppose V is a hypercube and suppose v is outward oriented for each face. Thus, mass is moving outside the volume. But the integral is negative since the inner product is positive (assuming v is not zero) and ρ is nonnegative. But mass is leaving the surface, so the left is clearly negative. Thus, the negative sign is needed.

In the next subsection, we will establish an argument in the sense of distributions.

2.2 Continuation equation definition in the weak sense

Let m be a vector-valued measure such that $dm = \rho dx$. Here, ρ is also vector-valued. Let ϕ be a C^∞ compactly supported test function. It is known by integration by parts in the theory of distributions

$$\int \phi d(\nabla \cdot m) = - \int \nabla \phi \cdot dm. \quad (13)$$

There is an equivalence

$$\int \phi(\nabla \cdot \rho) dx = - \int \nabla \phi \cdot \rho dx. \quad (14)$$

By the conservation of mass equation we saw on the previous slide, but in the weak form

$$\partial_t \int \phi \rho_s dx = - \int \phi(\nabla \cdot \rho_s v) dx. \quad (15)$$

Here, $\rho = \rho_s v$ is the vector field multiplied by scalar density.

It is a known fact from PDE theory

$$\partial_t \int \phi d\alpha - \int \nabla \phi \cdot v d\alpha = 0 = \partial_t \int \phi d\alpha - \int \phi \nabla \cdot v d\alpha = 0 \implies \partial_t \alpha + \nabla \cdot (v\alpha) = 0. \quad (16)$$

The second equality is due to integration by parts. This is because ϕ is any (smooth compactly supported) function. We will denote $d\alpha = \rho_s dx$. Thus,

$$0 = - \int \phi(\nabla \cdot \rho_s v) dx - \int \nabla \phi \cdot \rho dx \quad (17)$$

$$= \partial_t \int \phi \rho_s dx - \int \nabla \phi \cdot v \rho_s dx \quad (18)$$

$$\implies \partial_t \rho_s + \nabla \cdot (v \rho_s) = 0, \quad (19)$$

and we have the result.

2.3 Theorem deriving the continuity equation

Let $T_t : \mathbb{R}^n \rightarrow \mathbb{R}^n$, $0 \leq t \leq 1$, be a locally Lipschitz family of diffeomorphisms (isomorphism between manifolds) with $T_0 = \text{id}$. Let $v(t, x)$ be the velocity field associated with the trajectories $T_t = X(t, x)$,

$$\partial_t X(t, x) = v(t, X(t, x)). \quad (20)$$

Let α_0 be a probability distribution on $\Omega \subseteq \mathbb{R}^n$ and $\alpha(t, \cdot) = T_t \# \alpha_0$. Then $\alpha(t, \cdot)$ is the unique solution of the following linear transport equation:

$$\begin{cases} \partial_t \alpha + \nabla \cdot (\alpha v) = 0, 0 < t < 1 \\ \alpha(0, \cdot) = \alpha_0. \end{cases} \quad (21)$$

Proof. We previously saw the weak formulation of the continuity equation is such that

$$\partial_t \int \phi \rho_s dx = - \int \phi (\nabla \cdot \rho_s v) dx \quad (22)$$

in the previous section. Note that this formulation is the conservation of mass, which roughly follows from the definition of flux. This conservation of mass equation has heavy flavor from engineering fields, but we will mostly accept it as true. We will adopt $\alpha = \rho_s dx$ to be our measure with respect to Lebesgue measure, i.e. $\alpha(A) = \int_A \rho_s dx$, so ρ_s is a scalar density. Hence, if we can check for $\varphi \in C_0^\infty(\Omega)$

$$\partial_t \int \phi d\alpha = - \int \phi (\nabla \cdot v) d\alpha, \quad (23)$$

then we have the continuity equation holds. In particular, most of our proof will be attempting to boil down what is given to this form. We will begin with the equivalence Villani (chapter 1)

$$\int_{\Omega} \varphi(y) dT_t \# \alpha_0 = \int_{\Omega} \varphi(T_t(x)) d\alpha_0. \quad (24)$$

Note that it is not necessarily the case the domain need be the same in this formulation, but we will operate so that they are equal. Moreover, it is known

$$\frac{\partial}{\partial t}(\varphi \circ T_t) = \nabla \varphi(T_t) \cdot \frac{\partial T_t}{\partial t} = \nabla \varphi(T_t) \cdot v(t, T_t(x)). \quad (25)$$

The equivalence is by definition. As a reminder, this is a Euclidean inner product. Now, for all $h > 0$, we compute

$$\frac{1}{h} \left(\int_{\Omega} \varphi d \underbrace{\alpha(t+h, \cdot)}_{=T_{(t+h)} \# \alpha_0} - \int_{\Omega} \varphi d \underbrace{\alpha(t, \cdot)}_{T_t \# \alpha_0} \right) = \int_{\Omega} \frac{\varphi(T_{t+h}(x)) - \varphi(T_t(x))}{h} d\alpha_0, \quad (26)$$

which follows from 24. If we let $h \rightarrow 0$, then the above formulation is precisely a derivative, hence

$$\lim_{h \searrow 0} \int_{\Omega} \frac{\varphi(T_{t+h}(x)) - \varphi(T_t(x))}{h} d\alpha_0 = \frac{d}{dt} \int_{\Omega} \varphi d\alpha(t, \cdot), \quad (27)$$

where we have permitted regularity conditions. Moreover,

$$\frac{d}{dt} \int_{\Omega} \varphi d\alpha(t, \cdot) = \int_{\Omega} \nabla \varphi \cdot v d\alpha_0. \quad (28)$$

This follows from this equivalence. In particular,

$$\lim_{h \searrow 0} \int_{\Omega} \frac{\varphi(T_{t+h}(x)) - \varphi(T_t(x))}{h} d\alpha_0 = \int_{\Omega} (\partial_t \varphi(T_t(x))) d\alpha_0 = \int_{\Omega} \nabla \varphi(T_t) \cdot v(t, T_t(x)) d\alpha_0. \quad (29)$$

Note that is precisely conservation of mass.

There is more to prove. We will show uniqueness, i.e. if $\alpha_0 = 0$, then $\alpha(t, \cdot) = 0$ for all $0 < t < 1$. This shows uniqueness because we can take $\alpha_1 - \alpha_2 = 0$, and the result follows from well-posedness of the PDE. We examine the duality method. Assume that we can construct a function $\varphi(t, x)$ satisfying

$$\begin{cases} \frac{\partial \varphi}{\partial t} = -v \cdot \nabla \varphi \\ \varphi(s, \cdot) = \varphi_s, \quad \forall s < 1 \text{ fixed.} \end{cases} \quad (30)$$

Here, φ is an arbitrary test function

$$\frac{d}{dt} \int_{\Omega} \varphi(t, x) d\alpha(t, \cdot) \stackrel{\text{chain rule}}{=} \int_{\Omega} \frac{\partial \varphi}{\partial t} d\alpha(t, \cdot) + \int_{\Omega} \varphi(t, x) d \frac{\partial \alpha}{\partial t} \quad (31)$$

$$= \underbrace{\int_{\Omega} (-v \cdot \nabla \varphi) d\alpha(t, \cdot)}_{\text{follows by assumption}} + \underbrace{\int_{\Omega} \varphi(t, x) d(-\nabla \cdot (v\alpha))}_{\text{follows from continuity equation}} \quad (32)$$

$$= \int_{\Omega} (-v \cdot \nabla \varphi) d\alpha + \underbrace{\int_{\Omega} \nabla \varphi \cdot v d\alpha}_{\text{we refer to 13, which is an integration by parts for divergence of measures}} \quad (33)$$

$$= 0. \quad (34)$$

From this, we can conclude

$$\int_{\mathbb{R}^n} \varphi(s, x) d\alpha(s, \cdot) = \int_{\mathbb{R}^n} \varphi(0, x) d\alpha_0 = 0 \text{ since } \alpha_0 \text{ is the trivial measure,} \quad (35)$$

since the time derivative is zero, thus constant in time. Since φ was arbitrary, we conclude $\alpha(s, \cdot) = 0$, $0 \leq s \leq 1$.

From 30, we have using this and the chain rule

$$\partial_t \varphi + v \cdot \nabla \varphi = 0 \implies \frac{d}{dt} \varphi(t, X(t, x)) = 0 \quad (36)$$

$$\implies \varphi(t, T_t(x)) = \varphi(s, T_s(x)) \quad (37)$$

$$\implies \varphi(t, x) = \varphi(s, T_s \circ T_t^{-1}(x)). \quad (38)$$

□

Remark. If the trajectory has randomness, i.e.

$$dX(t, x) = v(t, X(t, x))dt + \sqrt{2\epsilon}dW \quad (39)$$

is obeyed, then the corresponding density follows

$$\partial_t \rho + \nabla \cdot (v\rho) = \epsilon \Delta \rho \quad (40)$$

$$\text{or equivalently} \quad (41)$$

$$\partial_t \rho + \sum_i \frac{\partial}{\partial x_i} (v_i \rho) = \epsilon \sum_i \frac{\partial^2}{\partial^2 x_i^2} \rho, \quad (42)$$

which is a Fokker-Planck equation with nonzero diffusion ($\epsilon \neq 0$).

3 The Benamou-Brenier formulation

If the Monge map exists, then the quadratic Wasserstein distance

$$W_2^2(\alpha_0, \alpha_1) = \min_{T \text{ such that } T\# \alpha_0 = \alpha_1} \int_{\Omega} \|x - T(x)\|^2 dx \quad (43)$$

has the equivalent formulation

$$A^2(\alpha_0, \alpha_1) = \begin{cases} \min_{\alpha, v} \iint_{[0,1] \times \mathbb{R}^n} \|v\|_2^2 d\alpha dt \\ \partial_t \alpha + \nabla \cdot (\alpha v) = 0 \\ \alpha(0, \cdot) = \alpha_0 \\ \alpha(1, \cdot) = \alpha_1. \end{cases} \quad (44)$$

Proof. Let us consider the class of functions

$$C(\alpha_0, \alpha_1) = \left\{ (\alpha, v) \left| \partial_t \alpha + \nabla \cdot (\alpha v) = 0, \alpha(0, \cdot) = \alpha_0, \alpha(1, \cdot) = \alpha_1 \right. \right\}. \quad (45)$$

Let $(\alpha, v) \in C(\alpha_0, \alpha_1)$. Consider the problem

$$\begin{cases} \frac{\partial X(t, x)}{\partial t} = v(t, X(t, x)) \\ X(0, x) = x, x \sim \alpha_0 \\ T_t : x \in \mathbb{R}^n \rightarrow X(t, x) \in \mathbb{R}^n \\ \alpha(t, \cdot) = T_t \# \alpha_0 \text{ satisfies } \partial_t \alpha + \nabla \cdot (\alpha v) = 0. \end{cases} \quad (46)$$

Thus, we have

$$\iint_{[0,1] \times \mathbb{R}^n} \|v(t, x)\|_2^2 d\alpha(t, \cdot) dt \quad (47)$$

$$\stackrel{\text{pushforward relation}}{=} \iint_{[0,1] \times \mathbb{R}^n} \|v(t, x)\|_2^2 dT_t \# \alpha_0 dt \quad (48)$$

$$\stackrel{\text{Villani}}{=} \iint_{[0,1] \times \mathbb{R}^n} \|v(t, T_t(x))\|_2^2 d\alpha_0 dt \quad (49)$$

$$= \iint_{[0,1] \times \mathbb{R}^n} \|v(t, X(t, x))\|_2^2 d\alpha_0 dt \quad (50)$$

$$\stackrel{\text{Fubini's (assume regularity)}}{=} \iint_{[0,1] \times \mathbb{R}^n} \|v(t, X(t, x))\|_2^2 dt d\alpha_0 \quad (51)$$

$$\stackrel{\text{Jensen's inequality}}{\geq} \int_{\mathbb{R}^n} \left\| \int_{[0,1]} v(t, X(t, x)) dt \right\|_2^2 d\alpha_0 \quad (52)$$

$$= \int_{\mathbb{R}^n} \left\| \int_{[0,1]} \frac{\partial X(t, x)}{\partial t} dt \right\|_2^2 d\alpha_0 \quad (53)$$

$$= \int_{\mathbb{R}^n} \|X(1, x) - X(0, x)\|_2^2 d\alpha_0 \quad (54)$$

$$= \int_{\mathbb{R}^n} \|T_1(x) - X(0, x)\|_2^2 d\alpha_0 \quad (55)$$

$$\stackrel{\text{since Wasserstein distance is optimal}}{\geq} W_2^2(\alpha_0, \alpha_1) \quad (56)$$

$$\implies A_2^2(\alpha_0, \alpha_1) \geq W_2^2(\alpha_0, \alpha_1). \quad (57)$$

Now, let $X(t, x) = x + t(T^*(x) - x)$, $v(t, x) = T^*(x) - x$, then the two inequalities become equality and we have the result. In particular, Jensen's inequality attains equality if the function to be integrated is independent of the variable integration (in other words, almost surely constant) [Pratt \(2014\)](#). Notice that $v(t, x) = T^*(x) - x$ is independent of time.

4 Calculus of variations review

Let $\mathcal{L}$ be a functional, typically such that

$$\mathcal{L} = \int_{\Omega} f dx. \quad (58)$$

Oftentimes, such as in Euler-Lagrange equation, $\Omega = [0, T]$ is a time-domain, and

$$\mathcal{L} = \int_{[0, T]} f(t, x(t), \dot{x}(t)) dt. \quad (59)$$

IF $\gamma : \Omega \rightarrow \mathbb{R}$, the first variation is

$$\delta \mathcal{L} = \left. \frac{d}{d\epsilon} \mathcal{L}(\gamma + \epsilon \psi) \right|_{\epsilon=0} = \lim_{\epsilon \rightarrow 0} \frac{\mathcal{L}(\gamma + \epsilon \psi) - \mathcal{L}(\gamma)}{\epsilon}. \quad (60)$$

for appropriate γ . Since $\mathcal{L}$ is the result of an integral, it would make sense that the first variation is also some real number and not a function.

The first variation is related to the Gateaux derivative. Let X, Y be locally convex Banach spaces, $U \subseteq X$ open, and $F : U \rightarrow Y$. The Gateaux differential DF of F at $u \in U$ in the direction of $\psi \in X$ is defined as

$$DF(u; \psi) = \left. \frac{d}{d\epsilon} F(u + \epsilon\psi) \right|_{\epsilon=0} = \lim_{\epsilon \rightarrow 0} \frac{F(u + \epsilon\psi) - F(u)}{\epsilon}. \quad (61)$$

From a practical point of view, the definition of the first variation is essentially the same as the Gateaux derivative. For our purposes, we will take $\mathbb{Y} \subseteq \mathbb{R}$. In particular, they are a bit different, since we will always take the functional output to be defined via integration, so it is a real number output, while the Gateaux derivative output is in Y . In fact, we can notice

$$DF : X \rightarrow Y. \quad (62)$$

To quote the first variation Wiki, the first variation "is recognizable as the Gateaux derivative of the functional." The Gateaux derivative is the generalization to the directional derivative but for differential calculus and for function spaces. Recall the directional derivative of $f(x)$ in the direction of v is

$$\nabla_v f(x) = \nabla_x f(x) \cdot v, \quad (63)$$

or equivalently,

$$\nabla_v f(x) = \lim_{\epsilon \rightarrow 0} \frac{f(x + \epsilon v) - f(x)}{\epsilon}. \quad (64)$$

This definition is without vector normalization with respect to v , consistent with the Gateaux derivative definition. These are equivalent because

$$\left. \frac{d}{d\epsilon} f(x + \epsilon v) \right|_{\epsilon=0} = \nabla f(x) \cdot v. \quad (65)$$

The functional derivative is the function $\partial \mathcal{L} / \partial \gamma$ such that

$$\left. \frac{d}{d\epsilon} \mathcal{L}(\gamma + \epsilon\psi) \right|_{\epsilon=0} = \iint \frac{\partial \mathcal{L}}{\partial \gamma}(x, t) \cdot \psi(x, t) dx dt. \quad (66)$$

Here, we are assuming the image of the functional derivative is in $\mathbb{R}$. In particular, the first variation is a real number. The functional derivative is a function corresponding to the first variation. In optimal transport, as in [Ansari](#), recall

$$\left. \frac{d}{d\epsilon} W_2^2(\gamma + \epsilon\phi; \nu) \right|_{\epsilon=0} = \int 2\psi_* d\phi = \int \frac{\delta W_2^2}{\delta \gamma} d\phi, \quad (67)$$

for appropriate ψ . Hence, ψ is the functional derivative. Note that $d\phi = \phi dx$ when considered absolute continuity w.r.t. Lebesgue measure. Recall a Wasserstein gradient flow is given by

$$\partial_t \mu_t - \nabla \cdot (\mu_t \nabla \frac{\delta F}{\delta \mu}(\mu_t)) = 0. \quad (68)$$

Hence, it is the functional derivative used in this definition, not the first variation.

Let us consider the example

$$\left. \frac{d}{d\epsilon} \mathcal{L}(\rho + \epsilon\psi) \right|_{\epsilon=0} = \frac{d}{d\epsilon} \iint (\partial_t(\rho + \epsilon\psi) + \operatorname{div}((\rho + \epsilon\psi)v)) \phi dx dt \quad (69)$$

$$= \frac{d}{d\epsilon} \iint (\partial_t \rho + \epsilon \partial_t \psi + \operatorname{div}(\rho v) + \epsilon \operatorname{div}(\psi v)) \phi dx dt \quad (70)$$

$$= \iint (\partial_t \psi + \operatorname{div}(\psi v)) \phi dx dt \quad (71)$$

$$= \int (\psi \phi|_{t=0}^{t=1} + \int (-\psi \partial_t \phi - \psi v \cdot \nabla \phi) dt) dx \quad (72)$$

$$= \iint (-\partial_t \phi - v \cdot \nabla \phi) \psi dt dx. \quad (73)$$

We have integration by parts and vanishing boundaries due to $n \cdot v = 0$. Thus the functional derivative is $\partial_t + v \cdot \nabla \phi$, which must be equal to 0. Notice the first variation is zero, and ψ is arbitrary. Note that ψ corresponds to what is admissible.

5 Coupling with the Hamilton-Jacobi equation

Consider the Lagrangian

$$\mathcal{L}(\rho, v; \phi, u) = \frac{1}{2} \iint_{[0,1] \times \Omega} \|v\|_2^2 \rho dx dt + \iint_{[0,1] \times \Omega} (\partial_t \rho + \operatorname{div}(\rho v)) \phi dx dt - \iint_{[0,1] \times \Omega} \rho u dx dt. \quad (74)$$

This is a weak formulation of Lagrange multipliers, where the typical multiplication with a Lagrange multiplier is modified into an analog with a product with a compactly supported function, and $u(t, x) \geq 0$. Now, notice

$$= \frac{1}{2} \iint_{[0,1] \times \Omega} \|v\|^2 \rho dx dt + \iint_{\Omega \times [0,1]} \partial_t \rho \phi dt dx + \iint_{[0,1] \times \Omega} \operatorname{div}(\rho v) \phi dx dt - \iint_{[0,1] \times \Omega} \rho u dx dt \quad (75)$$

$$\stackrel{\text{integration by parts}}{=} \frac{1}{2} \iint_{[0,1] \times \Omega} \|v\|^2 \rho dx dt + \int_{\Omega} (\phi \rho|_0^1 - \int_0^1 \rho \partial_t \phi) dx + \iint_{[0,1] \times \Omega} (-\rho v) \cdot \nabla \phi dx dt - \iint_{[0,1] \times \Omega} \rho u dx dt. \quad (76)$$

From standard Lagrange multiplier theory, we have a critical point at

$$\frac{\partial \mathcal{L}}{\partial \rho} = 0, \frac{\partial \mathcal{L}}{\partial v} = 0, \frac{\partial \mathcal{L}}{\partial \phi} = 0, \frac{\partial \mathcal{L}}{\partial u} = 0. \quad (77)$$

Taking such first variations, we will set these equal to zero. *Note:* in section 9.2, we saw a critical point to the optimization was done with respect to a constant, not zero. This was because our optimization in 9.2 was done w.r.t. ρ , so when the first variation was taken, this annihilated the term involving ρ , leaving the constant multiplier. In particular, this notion is consistent with the first variation is zero, since it is really just

$$\frac{\partial \mathcal{E}}{\partial \rho} - \lambda = 0, \quad (78)$$

before moving the constant to the other side. Returning to our proof, we get

$$\left\{ \begin{array}{l} \frac{\partial \mathcal{L}}{\partial \rho} = \frac{1}{2} \|v\|_2^2 - \phi_t - v \cdot \nabla \phi - u = 0 \iff \frac{1}{2} \|v\|_2^2 - \phi_t - v \cdot \nabla \phi = u \geq 0 \\ \frac{\partial \mathcal{L}}{\partial v} = \rho v - \rho \nabla \phi = 0 \stackrel{\rho > 0}{\implies} v = \nabla \phi \\ \partial_t \rho + \operatorname{div}(\rho v) = 0 \\ \rho(0, x) = \rho_0, \rho(1, x) = \rho_1 \\ v \cdot n(t, x) = 0. \end{array} \right. \quad (79)$$

If $\rho > 0$,

$$\left\{ \begin{array}{l} \frac{1}{2} \|\nabla \phi\|_2^2 - \phi_t - \nabla \phi \cdot \nabla \phi = 0 \implies \boxed{\partial_t \phi + \frac{1}{2} \|\nabla \phi\|_2^2 = 0} \\ \partial_t \rho + \operatorname{div}(\rho \nabla \phi) = 0 \\ \rho(0, \cdot) = \rho_0 \\ \rho(1, \cdot) = \rho_1 \\ v \cdot n = 0, x \in \partial \Omega. \end{array} \right. \quad (80)$$

The boxed equation is the Hamilton-Jacobi equation. In general, a Hamilton-Jacobi equation is any equation of the form

$$-\partial_t S = H(x, \nabla_x S, t), \quad (81)$$

which is the form our equation that is boxed is.

Let us denote $m(t, x) = p(t, x)v(t, x)$. Our optimization problem is equivalently

$$\left\{ \begin{array}{l} \min_{\rho, m} \frac{1}{2} \iint_{[0,1] \times \Omega} \frac{\|m(t, x)\|_2^2}{\rho(t, x)} dx dt \\ \text{such that} \\ \partial_t \rho(t, x) + \operatorname{div}(m(t, x)) = 0 \\ \rho(0, x) = \rho_0(x), \rho(1, x) = \rho_1(x) \\ m \cdot n(t, x) = 0, x \in \partial \Omega. \end{array} \right. \quad (82)$$

The integral equivalence is due to $\|m\|_2^2 = |\rho| \|v\|_2^2 = \rho \|v\|_2^2$, which is precisely the integrand of our original optimization (aside from the Lagrange multiplier terms). Note that last condition $m \cdot n$ is equivalent to $v \cdot n$ only because it is equal to zero. If it was nonzero, it would not be the same because the weighting ρ is a scalar multiplier that affects the weighting. The argument is similar to how a compactly supported test function has annihilating properties, because the density ρ can "scale v however across its domain." We remark ρ is not allowed to be anything, so the argument is different, but it has some similarities to compactly supported test arguments.

We will define the function

$$\theta(a, b) = \begin{cases} \|b\|_2^2 / a & a > 0 \\ 0, & a = b = 0 \\ +\infty, & a = 0, b \neq 0. \end{cases} \quad (83)$$

Note that θ is convex (it is basically a quadratic with discontinuities addressed) and lower semicontinuous (we will not give the exact definition, but not that it basically means "the lower half of all discontinuous parts contain all limit points"), i.e.

$$\liminf_{x \rightarrow x_0} f(x) \geq f(x_0). \quad (84)$$

For example, consider a jump discontinuity with an open point above and a closed point below. The $\liminf$ is the lower point, and so is $f(x_0)$ in this scenario.

Let us consider a transformation

$$\theta : p(t, x)m(t, x) \in \mathbb{R}^d \rightarrow \theta(p(t, x), m(t, x)) \in \mathbb{R}_{\geq 0}. \quad (85)$$

Due to the definition of θ , we arrive at the same optimization problem of 82 but rewritten with the double integral

$$\min_{\rho, m} \frac{1}{2} \iint_{[0,1] \times \Omega} \theta(p(t, x), m(t, x)) dx dt. \quad (86)$$

TO DO: prove θ is lower semicontinuous and convex. Since θ is convex, it can be noted that any optimization problem of the form

$$\min_{\rho_1, \dots, \rho_n} \iint \dots \int \Theta(\rho_1(t, x), \dots, \rho_n(t, x)) dx dt \quad (87)$$

has local minima are global minima if Θ is convex. It does not imply uniqueness, and see Gigili (2016) for a basic $\mathbb{R} \rightarrow \mathbb{R}$ example with infinitely many solutions (but they are all the same minima!).

6 The proximal operator and its gradient descent

Let us discuss proximal gradient descent and its qualities before returning to our optimization discussion. Proximal gradient descent is the iterative algorithm

$$x^{k+1} = \text{Prox}_{\tau g} \left(x^k - \tau \nabla f(x^k) \right), \quad \text{Prox}_{\tau g}(x) := \arg \min_u \left\{ \tau g(u) + \frac{1}{2} \|u - x\|_2^2 \right\}. \quad (88)$$

The goal of this method is that it satisfies the optimization problem

$$\min_x f(x) + g(x) \quad (89)$$

where

- f is convex, smooth with Lipschitz constant L
- g convex but not necessarily smooth .

In particular,

$$x^{k+1} = \text{Prox}_{\tau g} \left(x^k - \tau \nabla f(x^k) \right) = \arg \min_u \left\{ \tau g(u) + \frac{1}{2} \|u - x^k + \tau \nabla f(x^k)\|_2^2 \right\}. \quad (90)$$

In particular, by differentiating

$$\nabla_u \left\{ \tau g(u) + \frac{1}{2} \|u - x^k + \tau \nabla f(x^k)\|_2^2 \right\} = \tau \partial g(u) + (u - x^k + \tau \nabla f(x^k)) = 0 \quad (91)$$

Taking $u = x^{k+1}$ since u solves the arg min problem,

$$\tau \partial g(x^{k+1}) + (x^{k+1} - x^k + \tau \nabla f(x^k)) = 0, \quad (92)$$

or

$$x^{k+1} = x^k - \tau(\partial g(x^{k+1}) + \nabla f(x^k)), \quad (93)$$

meaning this algorithm does indeed minimize what is desired.

6.1 Convex function review

We will discuss what it means to be a convex a function. The following three definitions of convexity are equivalent: let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be a function defined over a convex set $U \subseteq \mathbb{R}^n$

- $f(\theta x + (1 - \theta)y) \leq \theta f(x) + (1 - \theta)f(y)$ (the chord is above the point along the graph)
- $f(y) \geq f(x) + \langle \nabla f(x), y - x \rangle$ (increasing)
- $\text{Hess}(f) \geq 0 \ \forall x$ (curvature condition).

Proof. 3 implies 2 is obvious, since

$$f(y) = f(x) + \langle \nabla f(x), y - x \rangle + (y - x)^T \nabla^2 f(\xi)(y - x) \quad (94)$$

for some ξ in the line segment between x and y , $\xi = \alpha x + (1 - \alpha)y$. 2 implies 3 is a little less trivial, due to the Hessian in the Taylor expansion is not evaluated at either x, y but rather a sufficient intermediary point. Since $x, y \in U$ are allowed to be chosen freely, this forces $\xi \in U$ to be arbitrary in U , thus the Hessian condition must hold for all (assuming sufficient smoothness) in U . It remains to show (1) implies (2) and vice versa [CS](#). Let us show (1) implies (2) first. Observe we can use the directional derivative and we get

$$\langle \nabla f(x), y - x \rangle = \lim_{h \rightarrow 0} \frac{f(x + h(y - x)) - f(x)}{h} \quad (95)$$

$$= \lim_{h \rightarrow 0} \frac{f((1 - h)x + hf(y)) - f(x)}{h} \quad (96)$$

$$\leq \lim_{h \rightarrow 0} \frac{(1 - h)f(x) + hf(y) - f(x)}{h} \quad (97)$$

$$= \lim_{h \rightarrow 0} \frac{h(f(y) - f(x))}{h} = f(y) - f(x). \quad (98)$$

(2) implies (1). Notice our (2) definition is equivalent to

$$f(y) + \langle \nabla f(x), x - y \rangle \geq f(x). \quad (99)$$

Choosing $x \leftarrow \theta x + (1 - \theta)y, y \leftarrow x$

$$f(\theta x + (1 - \theta)y) \leq f(x) + \langle \nabla f(\theta x + (1 - \theta)y), \theta x + (1 - \theta)y - x \rangle \quad (100)$$

$$= f(x) + (1 - \theta) \langle \nabla f(\theta x + (1 - \theta)y), y - x \rangle. \quad (101)$$

Following a similar argument, but choosing $x \leftarrow \theta x + (1 - \theta)y, y \leftarrow y$,

$$f(\theta x + (1 - \theta)y) \leq f(y) + \theta \langle \nabla f(\theta x + (1 - \theta)y), x - y \rangle. \quad (102)$$

Now, we get

$$f(\theta x + (1 - \theta)y) \quad (103)$$

$$= \theta f(\theta x + (1 - \theta)y) + (1 - \theta)f(\theta x + (1 - \theta)y) \quad (104)$$

$$\leq \theta f(x) + (1 - \theta)f(y) + \theta(1 - \theta) \langle \dots y - x \rangle + \theta(1 - \theta) \langle \dots x - y \rangle \quad (105)$$

$$= \theta f(x) + (1 - \theta)f(y) \quad (106)$$

as desired.

6.2 Subdifferential review

If $g : U \rightarrow \mathbb{R}$ is a real-valued convex function defined on a convex open set $U \subseteq \mathbb{R}^n$, we say v is in the subdifferential at x , $v \in \partial g(x)$, if for all $y \in U$,

$$g(y) \geq g(x) + \langle v, y - x \rangle \quad \forall y. \quad (107)$$

The subdifferential is always a nonempty convex compact set. In particular, note that we only need existence of g for the above to hold, not existence of the derivatives (i.e. the derivatives do not need to exist). Our set U is typically chose to be an open neighborhood, such an ϵ -ball, around the point of non-differentiability.

In the case that the function is real-valued, the subderivative will always be a closed interval. The subderivative interval boundaries will never exceed the two values in which the derivative exists in the open sets surrounding the set (possibly of zero measure) in which the subderivative is defined. Take for example $f = |x|$. The derivative here is ± 1 , and so the subderivative is $[-1, 1]$. In particular, Let us examine

$$|y| \geq |x| + v \cdot (y - x) \quad (108)$$

on an open neighborhood U around $0 \in \mathbb{R}$. Clearly we must take $x = 0$ at the cusp, and let us examine $y = y_0 > 0$. Hence,

$$y_0 \geq v y_0 \implies 1 \geq v. \quad (109)$$

So this is why the upper boundary of the subdifferential is 1. A similar argument falls in the negative case.

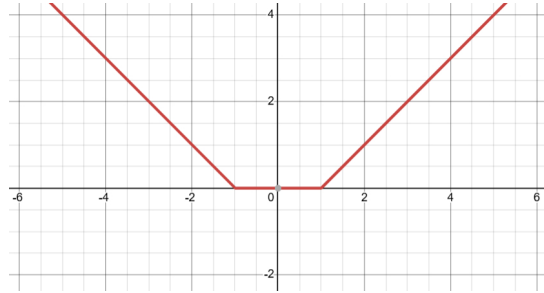


Figure 1: This figures illustrates two phenomenon: (1) there exist two points that are not differentiable but in which the subderivative exists; (2) this function is convex but infinitely many minima.

A subdifferential is an extension of the derivative for any function which is not typically differentiable but convex. For example, the subdifferential of $f(x) = |x|$ at $x = 0$ is $[-1, 1]$, meaning these are all possible "slopes" at $x = 0$ satisfying the subdifferential criterion. It can be noted ∂g is monotone, i.e. $\langle u_x - u_y, x - y \rangle \geq 0$, $u_x \in \partial g(x)$, $u_y \in \partial g(y)$. The proof of this following from

$$g(y) \geq g(x) + \langle u_x, y - x \rangle \quad (110)$$

$$g(x) \geq g(y) + \langle u_y, x - y \rangle \quad (111)$$

$$\implies g(y) + g(x) \geq g(x) + g(y) + \langle u_x, y - x \rangle + \langle u_y, x - y \rangle \quad (112)$$

$$\implies 0 \geq \langle u_x, y - x \rangle + \langle u_y, x - y \rangle = \langle u_x, y - x \rangle - \langle u_y, y - x \rangle \quad (113)$$

$$\implies 0 \leq \langle u_x - u_y, x - y \rangle. \quad (114)$$

Consider the example $f(x) = -x^2$. Take $u_x = \nabla f(1) = -2$, $u_y = \nabla f(0) = 0$. Then we get $\langle u_x - u_y, x - y \rangle = -2$, so it is not monotone, i.e. slope is decreasing (not convex).

6.3 Proximal continuation

Now we will show Prox is contractive. Note: contractive properties are very useful in optimization analysis because a contractive property can be used to prove something is convergent, i.e. $\|f(x^{k+1}) - f(x^k)\| \leq c^k \|x_0 - x^k\|$ where $c < 1$. Clearly the right-hand side will tend to 0 if $c < 1$ and the norm is finite. We will prove

$$\|\text{Prox}_g(x) - \text{Prox}_g(y)\| \leq \|x - y\|. \quad (115)$$

Proof. Consider

$$x' = \text{Prox}_g(x) = \arg \min_u g(u) + \frac{1}{2} \|u - x\|_2^2 \quad (116)$$

$$\implies 0 \in \partial g(x) + x' - x = \{y + (x' - x) : y \in \partial g(x)\} \quad (117)$$

$$\implies x - x' \in \partial g(x'). \quad (118)$$

Similarly, we can repeat this argument and get $y - y' \in \partial g(y')$. It follows from the monotone property

$$\langle u_{x, u_{x \in \partial g(x')} - u_{y, u_{y \in \partial g(y')}}}, x' - y' \rangle \geq 0 \quad (119)$$

$$\implies \langle (x - x') - (y - y'), x' - y' \rangle \geq 0 \quad (120)$$

$$\implies \|x - y\| \|x' - y'\| \stackrel{\text{Cauchy-Schwarz}}{\geq} \langle x - y, x' - y' \rangle \geq \|x' - y'\|^2 \quad (121)$$

$$\implies \|x - y\| \geq \|x' - y'\| = \|\text{Prox}_g(x) - \text{Prox}_g(y)\|, \quad (122)$$

giving the result.

□

Returning to proximal gradient descent,

$$x^{k+1} = \text{Prox}_{\tau g}(x^k - \tau \nabla f(x^k)) \quad (123)$$

$$= \text{Prox}_{\tau g}[(I - \tau \nabla f) \circ (x^k)]. \quad (124)$$

We use the $\circ$ notation to emphasize the above is well-defined, since I is a matrix and ∇f is a vector. Observe

$$x^* = \arg \min_x f(x) + g(x) \quad (125)$$

$$\implies 0 \in \tau \nabla f(x^*) + \tau \partial g(x^*) = \{\tau u_x + \tau \nabla f(x^*) : u_x \text{ satisfies } g(x^*) \geq \langle u_x, x^* - x \rangle\} \quad (126)$$

$$\in \tau \nabla f(x^*) + \tau(\text{anything in the subdifferential}) \quad (127)$$

$$\implies x^* - \tau \nabla f(x^*) \in x^* + \tau \partial g(x^*) \quad (128)$$

$$\implies (I - \tau \nabla f)(x^*) \in (I + \tau \partial g)(x^*) \quad (129)$$

$$\implies x^* \in \underbrace{(I + \tau \partial g)^{-1}}_{=\text{Prox}_{\tau g}} \circ (I - \tau \nabla f)(x^*) = \text{Prox}_{\tau g}((I - \tau \nabla f)(x^*)). \quad (130)$$

In particular, this shows the fixed point of proximal gradient descent yields the minimizer. Let us justify more why $(I + \tau \partial g)^{-1} = \text{Prox}_{\tau g}$.

Let $x = \text{Prox}_g(z)$. Now,

$$x = \text{Prox}_g(z) = \arg \min_u g(u) + \frac{1}{2} \|u - z\|_2^2 \implies \partial g(x) + \frac{1}{2} \partial_x \|x - z\|_2^2 = 0. \quad (131)$$

From this,

$$0 \in \partial g(x) + (x - z) \implies z \in x + \partial g(x) \implies x \in (I + \partial g)^{-1}(z). \quad (132)$$

Example 1. Consider the example $g(x) = |x| := \sum_i |x_i|$. Also,

$$\min_x \frac{1}{2} \|Ax - b\|^2 + \mu |x| \quad (133)$$

$$\text{Prox}_{\tau g}(y) = \arg \min_x \tau \mu |x| + \frac{1}{2} \|x - y\|^2 \quad (134)$$

$$= \left\{ \text{sign}(y_i) \cdot \max \left\{ |y_i| - \tau\mu, 0 \right\} \right\}_i. \quad (135)$$

This is the soft thresholding operator, and can be found by taking the partial ∂_{x_i} with respect to $\tau\mu x_i + \frac{1}{2}(x_i - y_i)^2$ and setting to 0.. In particular,

$$\begin{cases} x_i > 0 : & 0 = \tau\mu(1) + x_i - y_i \Rightarrow x_i = y_i - \tau\mu, & y_i > \tau\mu, \\ x_i < 0 : & 0 = \tau\mu(-1) + x_i - y_i \Rightarrow x_i = y_i + \tau\mu, & y_i < -\tau\mu, \\ x_i = 0 : & 0 \in \tau\mu[-1, 1] - y_i \Rightarrow |y_i| \leq \tau\mu. \end{cases} \quad (136)$$

Hence,

$$x_i = \text{sign}(y_i) \max\{|y_i| - \tau\mu, 0\}. \quad (137)$$

Also,

$$x^{k+1} = \text{Prox}_{\tau g} \left(x^k - \underbrace{\tau A^T(Ax^k - b)}_{=\tau \nabla f(x^k) = \tau \nabla \frac{1}{2} \|Ax^k - b\|^2} \right). \quad (138)$$

Example 2. Consider

$$\min_x \frac{1}{2} \|Ax - b\|^2 \text{ such that } x \in \Omega. \quad (139)$$

Here, $\Omega \subseteq \mathbb{R}^d$ is convex. Let us define

$$\chi_\Omega(x) = \begin{cases} \infty, & x \notin \Omega \\ 0, & x \in \Omega. \end{cases} \quad (140)$$

Our optimization problem can be rewritten to ensure the function is infinite off the domain with

$$\min_{x \in \mathbb{R}^d} \underbrace{\frac{1}{2} \|Ax - b\|^2}_{f(x)} + \underbrace{\chi_\Omega(x)}_{g(x)}. \quad (141)$$

We can check these functions satisfy the criteria for f, g established earlier. In particular, f is smooth, convex, Lipschitz, and g is convex but not smooth. Now,

$$\text{Prox}_{\tau g}(y) = \arg \min_x \chi_\Omega(x) + \frac{1}{2} \|x - y\|^2 \quad (142)$$

$$= \arg \min_{x \in \Omega} \frac{1}{2} \|x - y\|^2 \quad (143)$$

$$= \text{Proj}_\Omega(y), \quad (144)$$

and so

$$x^{k+1} = \text{Proj}_\Omega(x^k - A^T(Ax^k - b)). \quad (145)$$

6.4 Proximal algorithms in OT

Let us return to the OT problem we saw earlier. We established the problem

$$\begin{cases} \min_{\rho, m} \frac{1}{2} \iint_{[0,1] \times \Omega} (||m||^2 / \rho) dx dt \\ \text{such that} \\ C(\rho_0, \rho_1) := \begin{cases} \partial_t \rho + \text{div}(m) = 0 \\ \rho(0, x) = \rho_0 \\ \rho(1, x) = \rho_1 \\ m \cdot n = 0, x \in \partial\Omega. \end{cases} \end{cases} \quad (146)$$

Subject to constraints, our OT problem can be written using a similar χ function,

$$\min_{\rho, m} \frac{1}{2} \iint_{[0,1] \times \Omega} \frac{\|m\|^2}{\rho \geq 0} dxdt + \chi_{C(\rho_0, \rho_1)}(m, \rho). \quad (147)$$

Let us begin by simplifying this example. Let us consider a linearly constrained problem

$$\min_x f(x) + \chi_{Ax=b}(x) \iff \min_{Ax=b} f(x). \quad (148)$$

Applying proximal gradient descent (recall gradient descent will attempt to find a minima),

$$x^{k+1} = \text{Prox}_{\tau \chi_{Ax=b}}(\underbrace{x^k - \tau \nabla f(x^k)}_{=y^k}) \quad (149)$$

$$= \arg \min_u \tau \chi_{Au=b}(u) + \frac{1}{2} \|u - y^k\|_2^2 \quad (150)$$

$$= \arg \min_u \frac{1}{2} \|u - y^k\|_2^2 \text{ such that } Au = b. \quad (151)$$

Let us consider the method of Lagrange multipliers,

$$\mathcal{L}(u; \phi) = \frac{1}{2} \|u - y^k\|_2^2 + \langle b - Au, \phi \rangle \quad (152)$$

$$0 = \frac{\partial \mathcal{L}}{\partial u} = u - y^k - A^T \phi \quad (153)$$

$$0 = \frac{\partial \mathcal{L}}{\partial \phi} = b - Au. \quad (154)$$

Let us multiply the first equation equal to zero by A , and we will substitute in the second. As a remark, we assume A is wide so that AA^T can have an inverse (if it is tall, it cannot, since it would involve a mapping from a low to high dimension which can never be full rank for linear transformations). Notice

$$\underbrace{Au}_{=b} - Ay^k - AA^T \phi = 0 \quad (155)$$

$$\implies \phi = (AA^T)^{-1}(b - Ay^k) \quad (156)$$

$$\implies Au = Ay^k + AA^T(AA^T)^{-1}(b - Ay^k) \quad (157)$$

$$\implies u = y^k + A^T(AA^T)^{-1}(b - Ay^k). \quad (158)$$

In particular, "removing A " is justified. Note that A is not injective as it has nontrivial kernel since A is wide. Note that we do not have surjective is possible, since A maps from a high to low (or equal) dimension. The reason "removing A " is justified is by use of the first variation condition $u - y^k - A^T \phi$. Note in general taking away A cannot always be done without injectivity.

Let us return to OT. Let us consider the proximal gradient descent algorithm:

$$\begin{cases} m^{k+\frac{1}{2}} = m^k - \tau \frac{m^k}{\rho^k} \\ \rho^{k+\frac{1}{2}} = \rho^k + \tau \cdot \frac{1}{2} \cdot \frac{\|m^k\|^2}{(\rho^k)^2} \end{cases} \quad (159)$$

Here, we have used a $1/2$ step size, which is because the algorithm is coupled. This algorithm has linear complexity, and potentially it is greater without the $1/2$ step size. The two gradient descent conditions come from

$$\delta_m \int \frac{\|m\|^2}{\rho} dxdt \Rightarrow 2 \frac{m}{\rho} \quad (160)$$

$$\delta_\rho \int \frac{\|m\|^2}{\rho} dxdt \Rightarrow -\frac{\|m\|^2}{\rho^2}. \quad (161)$$

In particular, the right-hand side is a functional derivative calculation. The reason this works is minimizing via the first variation, carried through the functional derivative, will move towards minimizing the function. Recall a critical point of a functional will mean the first variation is zero. We will derive only the first one for brevity. Consider the definition of first variation:

$$\frac{d}{d\epsilon} F[m + \epsilon\nu; \rho] \Big|_{\epsilon=0} \quad (162)$$

$$= \frac{d}{d\epsilon} \Big|_{\epsilon=0} \iint \frac{\|m + \epsilon\nu\|^2}{\rho} dx dt \quad (163)$$

$$= \frac{d}{d\epsilon} \Big|_{\epsilon=0} \iint \frac{\langle m + \epsilon\nu, m + \epsilon\nu \rangle}{\rho} dx dt \quad (164)$$

$$= \iint \frac{2\langle m, \nu \rangle}{\rho} dx dt. \quad (165)$$

Note our functional derivative is therefore $2m/\rho$. Note that this is an inner product formulation for functional derivatives, which is valid when the function is vector-valued.

Let us consider a functional approach,

$$(m^{k+1}, \rho^{k+1}) = \operatorname{argmin}_{m, \rho} \left\{ \frac{1}{2} \iint_{[0,1] \times \Omega} (\rho - \rho^{k+\frac{1}{2}})^2 dx dt + \frac{1}{2} \iint_{[0,1] \times \Omega} \|m - m^{k+\frac{1}{2}}\|_2^2 dx dt \right\} \quad (166)$$

$$= \operatorname{argmin}_{m, \rho} \left\{ \frac{1}{2} \|\rho - \rho^{k+\frac{1}{2}}\|_{L^2}^2 + \frac{1}{2} \|m - m^{k+\frac{1}{2}}\|_{L^2}^2 \right\}, \quad (167)$$

where an L^2 norm is defined using the typical L^p integral norm on a vector-valued function f with respect to a measure μ

$$\|f\|_{L^p} := \left(\int \|f\|^p d\mu \right)^{1/p}, \quad (168)$$

such that the conditions hold:

$$\begin{cases} \frac{\partial \rho}{\partial t} + \operatorname{div} m = 0 \\ \rho(0, x) = \rho_0 \\ \rho(1, x) = \rho_1 \\ m \cdot n = 0 \quad x \in \partial\Omega. \end{cases} \quad (169)$$

Consider a Lagrange multipliers approach to solving this optimization problem

$$L(\rho, m; \phi) = \frac{1}{2} \|\rho - \rho^{k+\frac{1}{2}}\|_{L^2}^2 + \frac{1}{2} \|m - m^{k+\frac{1}{2}}\|_{L^2}^2 + \iint_{[0,1] \times \Omega} \phi \left(\frac{\partial \rho}{\partial t} + \nabla \cdot m \right) dx dt \quad (170)$$

$$= \frac{1}{2} \|\rho - \rho^{k+\frac{1}{2}}\|_{L^2}^2 + \frac{1}{2} \|m - m^{k+\frac{1}{2}}\|_{L^2}^2 \quad (171)$$

$$- \iint_{[0,1] \times \Omega} \rho \frac{\partial \phi}{\partial t} dx dt + \int_{\Omega} \phi(1, \cdot) \rho_0 dx - \int_{\Omega} \phi(0, \cdot) \rho_0 dx - \iint_{[0,1] \times \Omega} m \cdot \nabla \phi dx dt \quad (172)$$

Here, we have used integration by parts. Now, we have done several functional derivative calculations at this point, so we will omit the following calculations. As a remark, recall the L^2 norm is an integral norm. Computing the functional derivatives,

$$0 = \frac{\partial L}{\partial \rho} = \rho - \rho^{k+1/2} - \frac{\partial \phi}{\partial t} \Rightarrow \rho = \rho^{k+1/2} + \frac{\partial \phi}{\partial t} \quad (173)$$

$$0 = \frac{\partial L}{\partial m} = m - m^{k+1/2} - \nabla \phi \Rightarrow m = m^{k+1/2} + \nabla \phi \quad (174)$$

$$0 = \frac{\partial L}{\partial \phi} = \frac{\partial \rho}{\partial t} + \nabla \cdot m = 0 \quad (175)$$

Recall m is vector-valued, ϕ, ρ are scalar-valued. Substituting the first two into the third,

$$\frac{\partial \rho^{k+1/2}}{\partial t} + \frac{\partial^2 \phi}{\partial t^2} + \nabla \cdot (m^{k+1/2} + \nabla \phi) = 0, \quad (176)$$

which gives

$$\begin{cases} -\frac{\partial^2 \phi}{\partial t^2} - \Delta \phi = \frac{\partial p^{k+1/2}}{\partial t} + \nabla \cdot m^{k+1/2} \\ \partial_t \phi(t, x) = 0 \quad \text{for } t = 0, 1, \quad x \in \Omega \\ \nabla \phi(t, x) \cdot n = 0 \quad \forall t, \quad x \in \partial\Omega. \end{cases} \quad (177)$$

In particular, it is known $m \cdot n = 0$, and it is also true $m = m^{k+1/2} + \nabla \phi$. If $m \cdot n = 0$ holds for all m increments, we get the last equation. Similarly, the boundary conditions on $\rho(0, x) = \rho_0, \rho(1, x) = \rho_1$, and $\rho = \rho^{k+1/2} + \partial_t \phi$ gives the second condition. There exists a numerical method for the above equation. It can be solved in $\mathcal{O}(n \log n)$ or $\mathcal{O}(n)$ time. We will not go into this in detail, as it requires reading some papers.

Again using a functional derivative approach,

$$\begin{aligned} m^{k+1} &= m^{k+1/2} + \nabla \phi \\ p^{k+1} &= p^{k+1/2} + \frac{\partial \phi}{\partial t} \end{aligned}$$

This is slightly nontrivial, so we will derive the first one. Let us ignore terms not involving m , since those will just become 0. We will consider a vector-valued perturbation. Notice

$$\left. \frac{d}{d\epsilon} \right|_{\epsilon=0} L_M(\rho, m + \epsilon \nu; \phi) = \left. \frac{d}{d\epsilon} \right|_{\epsilon=0} \left(\frac{1}{2} \|m + \epsilon \nu - m^{k+1/2}\|_{L^2}^2 - \iint (m + \epsilon \nu) \cdot \nabla \phi dx dt \right) \quad (178)$$

$$= \left. \frac{d}{d\epsilon} \right|_{\epsilon=0} \left(\iint \frac{1}{2} \langle m + \epsilon \nu - m^{k+1/2}, m + \epsilon \nu - m^{k+1/2} \rangle dx dt \right) - \iint \nu \cdot \nabla \phi dx dt \quad (179)$$

$$= \left. \frac{d}{d\epsilon} \right|_{\epsilon=0} \left(\iint \langle \epsilon \nu, m - m^{k+1/2} \rangle + \text{terms not linear in } \epsilon \, dx dt \right) - \iint \nu \cdot \nabla \phi dx dt \quad (180)$$

$$= \left(\iint \langle \nu, m - m^{k+1/2} - \nabla \phi \rangle dx dt. \right) \quad (181)$$

Therefore $m^{k+1} = m^{k+1/2} + \nabla \phi$ is the solution to the first variation condition of zero.

7 Other optimization methods

7.1 The Schrödinger bridge

In this section, we consider the optimization problem

$$\min_{m, p} \iint_{[0,1] \times \Omega} \frac{\|m\|^2}{\rho} dx dt \quad (182)$$

where we have constrained

$$\text{s.t.} \quad \begin{cases} \frac{\partial p}{\partial t} + \text{div}(m) = \xi \Delta p \\ \rho(0, \cdot) = \rho_0 \\ \rho(1, \cdot) = \rho_1 \\ m \cdot n = 0 \quad \text{on } \partial\Omega. \end{cases} \quad (183)$$

In particular, there is now diffusion, as our Fokker-Planck PDE now has nonzero diffusion. In particular, the density experiences a smoothing effect that is not attributed to the drift vector field m .

7.2 The ADMM problem

Consider the problem

$$\min \frac{1}{2} \iint_{[0,1] \times \Omega} \frac{\|m\|^2}{\rho} + \chi_{C(\rho_0, \rho_1)}(m, \rho), \quad (184)$$

where

$$C(\rho_0, \rho_1) = \left\{ (m, \rho) \left| \begin{array}{l} \rho_t + \operatorname{div} m = 0, \rho(1, \cdot) = \rho_1, \rho(0, \cdot) = \rho_0, m \cdot n = 0 \end{array} \right. \text{ on } \partial\Omega \right\}. \quad (185)$$

As before, we have defined

$$\chi_{C(\rho_0, \rho_1)}(f) = \begin{cases} \infty, & f \notin C(\rho_0, \rho_1) \\ 0, & f \in C(\rho_0, \rho_1). \end{cases} \quad (186)$$

This can be rewritten in a variable splitting problem

$$\min_{(m, \tilde{m}, \rho, \tilde{\rho})} \frac{1}{2} \iint_{[0,1] \times \Omega} \frac{\|m\|^2}{\rho} + \chi_{C(\rho_0, \rho_1)}(\tilde{m}, \tilde{\rho}), \quad (187)$$

where

$$m = \tilde{m}, \rho = \tilde{\rho}. \quad (188)$$

8 JKO conditions with continuity equation equivalence

We show the JKO conditions as in [Lee et al. \(2023\)](#) are equivalent to a continuity equation.

Consider the density under the flow map $\rho(t, T(t, x))$. By the chain rule,

$$\frac{d}{dt} \rho(t, T(t, x)) = \partial_t \rho + \nabla \rho \cdot \partial_t T(t, x). \quad (189)$$

We are aware that $\partial_t T = v$, so that gives

$$\frac{d}{dt} \rho = \partial_t \rho + \nabla \rho \cdot v. \quad (190)$$

Now, consider the following:

$$\frac{d}{dt} \rho(t, T(t, x)) = \frac{d}{dt} \frac{\rho^k(x)}{\det|J(t, x)|} = - \frac{\rho^k(x)}{(\det|J(t, x)|)^2} \cdot \partial_t \det|J(t, x)| \quad (191)$$

$$= - \frac{\rho^k(x)}{\det|J(t, x)|} \cdot \frac{\partial_t \det|J(t, x)|}{\det|J(t, x)|} = - \frac{\rho^k(x)}{\det|J(t, x)|} \partial_t \log \det|J(t, x)| \quad (192)$$

$$= - \rho(t, T(t, x)) \partial_t \log \det|J(t, x)| = - \rho(t, T(t, x)) \operatorname{div}(v). \quad (193)$$

Thus,

$$- \rho(t, T(t, x)) \operatorname{div}(v) = \partial_t \rho + \nabla \rho \cdot v \implies \partial_t \rho + \nabla \cdot (\rho v) = 0, \quad (194)$$

as desired. The equivalence of $\partial_t \log \det|J(t, x)| = \operatorname{div}(v)$ is due to Jacobi's formula.

9 Derivation of the JKO algorithm

9.1 The Euclidean case

First, we discuss the Euclidean case.

Consider the optimization problem

$$\min_x \left\{ \frac{1}{2\tau} \|x - x_k\|_2^2 + \mathcal{E}(x) \right\}. \quad (195)$$

Note that the solution existence is dependent on conditions, such as convexity. A local minima to this problem can be found by taking the gradient

$$\nabla \left[\frac{1}{2\tau} \|x - x_k\|_2^2 + \mathcal{E}(x) \right] = \frac{1}{\tau} (x - x_k) + \nabla \mathcal{E}(x) = 0. \quad (196)$$

As a side remark, the first term derivative can be seen more clearly using

$$\nabla \langle x - x_k, x - x_k \rangle = [\partial_j (\sum_i (x_i - x_{k,i})^2)]_j = [2(x_j - x_{k,j})]_j. \quad (197)$$

Here, $[\cdot]_j$ denotes vector concatenation. Rearranging our equation, we see

$$\frac{x - x_k}{\tau} - \nabla \mathcal{E}(x), \quad (198)$$

which is exactly the Euler scheme for the gradient flow equation

$$\partial_t x(t) = -\nabla \mathcal{E}(x(t)). \quad (199)$$

9.2 The Wasserstein case

The Wasserstein gradient flow PDE is

$$\partial_t \rho = \nabla \cdot (\rho \nabla (\mathcal{E}'(\rho))). \quad (200)$$

We have denoted $\mathcal{E}'$ the first variation of the functional. Note that [Ansari](#) is a good reference for this section. Similarly as before, consider the optimization problem

$$\min_{\rho} \left\{ \frac{1}{2\tau} \int |x - T(x)| d\rho + \mathcal{E}(\rho) \right\}. \quad (201)$$

We cannot take the gradient of this optimization problem, as we did in the Euclidean case. Instead, the equivalent is the first variation, so we have a solution to this problem takes the form

$$\delta \left(\frac{1}{2\tau} \int |x - T(x)| d\rho + \mathcal{E}(\rho) \right) = \text{constant}. \quad (202)$$

The reason it is constant and not zero is because ρ is constrained, so the constant follows from the method of Lagrange multipliers since the first variation of

$$\delta \left(\lambda \left(\int \rho - 1 \right) \right) = \lambda. \quad (203)$$

It is known $T(x) = x - \nabla \phi(x)$ for some convex ϕ [Mokrov et al. \(2021\)](#) [Bao and Zhang \(2022\)](#) (page 4) [Ansari](#). It is also known the first variation of the Wasserstein distance is [Ansari](#)

$$\delta \left(\frac{1}{2\tau} W_2^2(\rho, \rho_k) \right) = \frac{1}{\tau} \phi. \quad (204)$$

Thus,

$$\frac{1}{\tau} \phi(x) + \mathcal{E}'(\rho_k)(x) = \text{constant} \implies \frac{1}{\tau} \nabla \phi(x) + \nabla \mathcal{E}'(\rho_k)(x) = 0. \quad (205)$$

Equivalently,

$$\frac{1}{\tau} (x - T(x)) = -\nabla \mathcal{E}'(\rho_k)(x). \quad (206)$$

In the continuum limit, we get

$$v = -\nabla \mathcal{E}'(\rho_k), \quad (207)$$

which yields our Wasserstein gradient flow when substituted into [200](#).

10 Wasserstein geometry

In this section, we discuss the Riemannian metric of Wasserstein space. Recall the Riemannian metric is defined as

$$g|_p : \text{Tan}_p M \times \text{Tan}_p M \rightarrow \mathbb{R}^*, g_{ij} = g(\partial_{x^i}|_p, \partial_{x^j}|_p). \quad (208)$$

We have used nonstandard notation here, and we will denote $\mathbb{R}^* = \mathbb{R}^+$ corresponding to $i = j$ and $\mathbb{R}^* = \mathbb{R}$ corresponding to $i \neq j$.

Consider the set of Borel measures on M such that [Takatsu](#)

$$\mathcal{P}_2(M) = \left\{ \mu : \text{Borel sets on } M \rightarrow \mathbb{R}^+ \cup \{0\} : \int_M d(x, x_0)^2 d\mu(x) < \infty, x_0 \in M, \int_M d\mu(x) = 1 \right\}. \quad (209)$$

We remark this notation is also a bit nonstandard, but is very clear, so we will use it. As a remark, recall that $M = \mathbb{R}^d$ is indeed a valid Riemannian manifold. Our continuity equation stems from this set.

The Riemannian metric on Wasserstein space is [Ay \(2024\)](#) [Craig and contributors](#)

$$\left\langle \frac{\partial \phi}{\partial t_1}, \frac{\partial \psi}{\partial t_1} \right\rangle_\mu = g_W(\nabla \phi, \nabla \psi) = \int_M \langle \nabla \phi, \nabla \psi \rangle d\mu(x). \quad (210)$$

Recall call that an induced metric is an inner product of tangent vectors, so the above formulation is consistent with such an inner product definition. Here, ϕ, ψ are solutions to a continuity equation

$$\partial_t \mu_t + \nabla \cdot (\mu_t \nabla \Psi) = 0. \quad (211)$$

The inner product is in such a way that ϕ, ψ are tangent vectors located at μ . Such holds in the sense of distributions [Ambrosio et al. \(2008\)](#)

$$\iint_{[0,1] \times \mathbb{R}^d} (\partial_t \varphi + \langle v_t, \nabla \varphi \rangle) d\mu_t dt = 0, \quad \varphi \text{ is a suitable compactly supported test function}, \quad (212)$$

where the divergence is defined in the sense of distributions (or we can define the gradient of the measure using the Radon-Nikodym derivative, which is the density). Note that the tangent bundle is

$$\text{Tan}_\mu \mathcal{P}(M) := \overline{\left\{ \nabla \varphi : \varphi \in \mathcal{P}_2(M) \right\}}^{L^2(\mu; X)}. \quad (213)$$

Also, our vector field is such that

$$v_t \in \left\{ \nabla \Psi : \Psi \text{ is a plausible potential} \right\} = \Gamma, \Gamma^* = \overline{\left\{ \nabla \Psi : \Psi \in \mathcal{C}(M) \right\}}^{L^p(\mu; X)}. \quad (214)$$

Γ is a generalized set of admissible potentials, and Γ^* is a more prototypical example. Here, the closure is taken to ensure the set is compact. $\mathcal{C}$ is a set of admissible smooth functions. We refer to [Ambrosio et al. \(2008\)](#), chapter 8 for discussion of this. In particular, it is known (page [Ambrosio et al. \(2008\)](#))

$$v_t \in \overline{\left\{ j_q(\nabla \Psi) : \Psi \in \text{Cyl}(X) \right\}}^{L_p(\mu_t; X)} \quad (215)$$

and j_q is the map $j_q(v) = \|v\|_2^{q-2} v$. Here Cyl denotes the cylindrical class of functions

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